

Business Problem Statement — HR Analytics & Employee Attrition Dashboard

IBM's HR department is experiencing increasing employee attrition across multiple departments, leading to higher recruitment costs, workforce instability, and reduced organizational productivity. The company wants to better understand the major factors contributing to employee turnover, including overtime workload, salary levels, employee demographics, job satisfaction, and work-life balance.

Using the **IBM HR Analytics Employee Attrition & Performance Dataset**, the objective of this project is to analyze workforce trends, identify the key drivers of employee attrition, and support data-driven HR decision-making through interactive analytics and visualization.

The project aims to answer the following business questions:

- Which departments experience the highest employee attrition?
- How does overtime affect employee turnover?
- Which salary groups and age groups are most likely to leave the organization?
- How do job satisfaction and employee demographics influence attrition?

The final solution includes data cleaning in Excel, analytical querying using PostgreSQL, and an interactive HR analytics dashboard built in Power BI to generate actionable workforce insights.

Deliverables

1. Data Preparation & Cleaning (Excel)

- Cleaned and standardized employee data.
- Verified missing values, duplicates, and data consistency.
- Created additional fields such as Age Group and Salary Slab for better workforce segmentation.

2. Data Analysis using PostgreSQL

- Structured HR data into relational format.
- Performed SQL queries to analyze:
 - Attrition by department
 - Overtime impact on resignations
 - Salary-based attrition
 - Employee demographics

- Job satisfaction trends
- Workforce segmentation

3. Visualization & Dashboard Development (Power BI)

- Developed an interactive HR analytics dashboard with:
 - KPI cards
 - Attrition analysis
 - Salary insights
 - Gender-wise distribution
 - Department-level turnover analysis
 - Employee demographic insights
 - Interactive slicers and filters

4. Insights & Business Recommendations

- Identified key workforce attrition patterns.
- Generated HR recommendations to improve employee retention, engagement, and workforce planning.

5. GitHub Repository

- Maintained a structured repository containing:
 - Excel preprocessing files
 - PostgreSQL queries
 - Power BI dashboard
 - Project documentation